IMPLEMENTING TOPOLOGICAL DATA ANALYSIS TO OPTIMIZE FORECASTING OF SHORT-HORIZON VOLATILITY IN HIGH FREQUENCY CRYPTOCURRENCY MARKETS

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Abstract. This paper builds upon the novel introduction of Topological Data Analysis as a feature in machine learning models used to trade equities. Specifically, we aim to improve upon current industry-standard realized-volatility models within high-frequency cryptocurrency markets. We do this by adding various topological features derived from persistent homology, in an aim to improve forecasting performance during volatile periods. Using one-minute BTC/USDT and ETH/USDT data spanning 2020–2023, we embed rolling windows of returns as point clouds, compute persistence diagrams, and extract twelve topological summary features that augment both a linear (HAR-RV) and a nonlinear (XGBoost) benchmark. We identify and correct for a frequency confound: because topological features are computed on finer-frequency returns than the baseline models observe, part of their apparent benefit is attributable to frequency information rather than topology itself. After controlling for this with a ladder of robust one-minute volatility estimators, we find that the naive six-to-fourteen percent reduction in QLIKE loss shrinks by roughly half, but a statistically decisive residual gain of 3.2 to 5.9 percent survives (Diebold–Mariano statistics between 7.6 and 9.9). This residual gain concentrates in turbulent regimes and nonlinear models, and is carried in part by a scale-invariant topological feature. We therefore provide evidence that persistent homology contributes genuine, non-redundant information for forecasting realized volatility in high-frequency cryptocurrency markets during turbulent periods, provided the frequency confound is properly controlled.

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1. Introduction

At short horizons, it is extremely difficult to predict returns of financial assets such as equities, currencies, or cryptocurrencies. However, it is very common to see future volatility predicted with accuracy in the same settings (Andersen et al., 2003). Forecasts of short-horizon volatility are the baseline of risk management, position sizing, and options pricing. These forecasts are generally produced by statistical and machine learning models, with the industry standard being the Heterogeneous Autoregressive model of Corsi (Corsi, 2009). This model is a simple linear regression on backward-looking volatility averages that has proven remarkably difficult to beat. Naturally, cryptocurrency markets are an attractive place to start trying to beat these models, or at least improve on them. Continuous trading, published high-frequency data, and many volatile periods (May 2021 sell-off, Terra/Luna collapse, FTX failure), are the reasons for which we choose crypto markets to benchmark our improved models upon.

Models in the HAR family summarize the recent past through scalar volatility averages, and in doing so they discard a lot of information about the shape of returns. Using Topological Data Analysis, we attempt to recover some of that structure. After embedding a window of returns as a point cloud, building simplicial complexes upon it, and tracking the birth and death of topological features using filtration, we can use persistent homology to find out more information about our data. We can find out how its points cluster, whether the trajectory recurs upon itself, and how the geometric arrangement changes from one window to the next. A natural question to ask is how we can confirm that these summaries of data aren't just tracking noise, as is common in high-frequency settings. We set up stability theorems that confirm that small perturbations in data move the summaries only slightly, allowing us to use them as real features. A growing field in literature has been applying these tools to financial time series: Gidea and Katz (Gidea and Katz, 2018) showed that persistence landscapes of equity indices rise ahead of major crashes, and Souto (Souto, 2023) found that topological features improve realized-volatility forecasts on daily equity data, with the gains concentrated in turbulent periods and in nonlinear models. Souto termed this phenomenon "Topological Tail Dependence".

This paper tests whether the findings in past literature withstand a change of setting. We move from daily to high-frequency data as well as moving to cryptocurrency markets, whose thin venues and structural breaks are hostile to fragile methods. In doing so, we are able to confront a hazard that, to our knowledge, prior work in this literature has not addressed. Topological features computed from finer-frequency returns than the benchmark model observes necessarily carry information about that finer frequency. In fact, we find that our most important topological feature correlates at 0.83 (BTC) and 0.70 (ETH) with the log of one-minute realized variance over the trailing hour, a quantity our five-minute baselines never see. Thus, we need to make sure that any comparisons we make are truly testing topology,

rather than the finer-frequency information that it provides. Therefore, we construct a ladder of controls, augmenting the baselines with progressively stronger non-topological volatility estimators computed from the same one-minute returns, jump-robust bipower variation, and realized range. Then, we test the topological features that we built against the strongest combination of controls.

Our results, while slightly diminished by our stronger comparisons, still hold significantly. Naively, topological features reduce the QLIKE loss of linear and nonlinear baselines by 6 to 14 percent. However, we find that roughly half of that improvement is reproduced by the robust volatility controls alone. Our remainder (between 3.2 and 5.9 percent depending on asset and model) is statistically decisive (Diebold–Mariano statistics between 7.6 and 9.9). These findings concentrate in volatility spikes and turbulent regimes, and are carried in part by a specific topological feature that is invariant to the scale of returns. Ablations further confirm that our results are not due to the nonlinear model's wider set of features nor to any liquidity indicators.

The first half of our paper consists of an exposition of the mathematics behind our twelve persistent homology features, beginning with the creation of simplicial complexes from point clouds and diving into distances and vectorization of our data. This material rests on the foundational Sections 2 through 7, assuming only a basic understanding of linear algebra and analysis, so a reader new to the topic can follow the whole process, from definitions to the formulation of models. After the expository section, we describe the building of a causal, leakage-controlled pipeline that converts a stream of one-minute prices into topological features suitable for standard forecasting models, documenting and explaining design choices. We then identify the frequency confound described above and propose the control-ladder methodology for isolating topological content from frequency content. We believe this control should become standard practice when topological features enter forecasting comparisons.

2. Simplicial Complexes

Before we can study the shape of a finite set of data points, we need a way to build structure on it. The idea is to create shapes from collections of points and edges. These pieces are called *simplices*; a *geometric k-simplex* is the convex hull of $k+1$ affinely independent points $v_0, \dots, v_k \in \mathbb{R}^N$, so that a 0-simplex is a point, a 1-simplex an edge, a 2-simplex a filled triangle, and so on.

Definition 2.1 (Geometric k-simplex)**.** A *geometric k-simplex* σ is the smallest convex set containing $k+1$ affinely independent points $v_0, \dots, v_k \in \mathbb{R}^N$:

$$\sigma = \left\{ \sum_{i=0}^{k} t_i v_i \;\middle|\; t_i \geq 0,\ \sum_{i=0}^{k} t_i = 1 \right\}.$$

The points $v_0, \dots, v_k$ are its *vertices* and k is its *dimension*.

Definition 2.2 (Abstract simplicial complex)**.** Let V be a vertex set, not necessarily finite. An *abstract simplicial complex* K on V is a collection of nonempty finite subsets of V (*simplices*) such that:

(i) $\{v\} \in K$ for every $v \in V$;
(ii) if $\sigma \in K$ and $\emptyset \neq \tau \subseteq \sigma$, then $\tau \in K$.

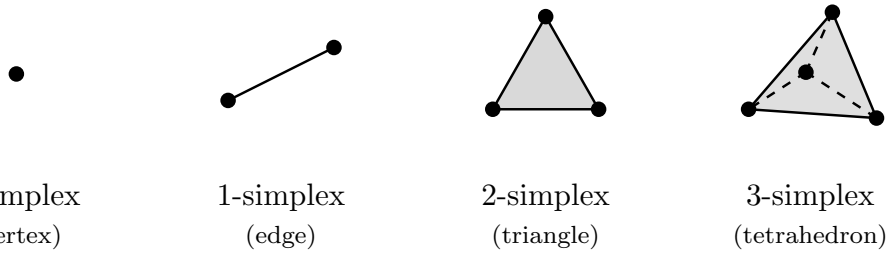


FIGURE 1. The simplices of dimension 0 through 3: a vertex, an edge, a filled triangle, and a solid tetrahedron. A k-simplex is spanned by $k+1$ affinely independent vertices.

A simplex with $k+1$ vertices is a *k-simplex*, and the *dimension* of K is the largest dimension of any simplex in K.

Although the definition permits an infinite vertex set, every complex in this paper is built on a finite point cloud, so V will always be finite for us. The geometric picture above is useful for intuition, but for computation it is easier to forget the exact coordinates and only record the collections of vertices that are filled in. Since we have downward closure, we know that if an abstract simplicial complex contains a filled-in shape, it also must contain all of its "sub-shapes." Below, we provide an example.

Example 2.3. Suppose we wish to fill in the triangle on the vertices $\{a, b, c\}$. Downward closure forces the complex to contain not only the 2-simplex $\{a, b, c\}$ but also its edges $\{a, b\}, \{b, c\}, \{a, c\}$ and its vertices $\{a\}, \{b\}, \{c\}$. Additionally, the collection consisting of $\{a, b\}, \{b, c\}, \{a, c\}, \{a\}, \{b\}, \{c\}$, but not the triangle, is also a valid simplicial complex. Geometrically, it is the hollow boundary of the triangle rather than the filled triangle.

Definition 2.4 (Geometric realization)**.** Let K be a simplicial complex on a finite vertex set V. Choose a map $\alpha : V \to \mathbb{R}^N$ such that for every simplex $\sigma = \{v_0, \ldots, v_k\} \in K$ the points $\alpha(v_0), \ldots, \alpha(v_k)$ are affinely independent, and write $|\sigma|_\alpha \subset \mathbb{R}^N$ for the convex hull they span (a geometric k-simplex). Suppose moreover that any two of these geometric simplices intersect precisely in the geometric simplex of their shared face, or not at all. Then the *geometric realization* of K is the union

$$|K|_\alpha = \bigcup_{\sigma \in K} |\sigma|_\alpha,$$

regarded as a topological space with the subspace topology from $\mathbb{R}^N$.

Such a map α always exists once N is large enough, and any two valid choices produce homeomorphic realizations, so we may speak of the geometric realization $|K|$. This process transforms an abstract set of points and connections into a subset of $\mathbb{R}^N$ that approximates the shape of the data. As we see below, by varying the threshold distance step by step, we can create a sequence of simplicial complexes (a filtration), which allows us to track how the shape of the data evolves across different scales. From here on, we will move freely between simplicial complexes and their geometric realizations.

3. Simplicial Homology

Next, we will describe how to measure the topological features of complexes. Homology converts a simplicial complex into a sequence of vector spaces whose dimensions count the holes in each shape. Then, through linear algebra, we can detect which cycles create these holes.

Notation 3.1. Throughout this section $\mathbb{F}$ denotes a fixed field. In this paper we will take $\mathbb{F} = \mathbb{Z}_2 = \{0, 1\}$. Recall that in $\mathbb{Z}_2$, $-1 = 1$.

Definition 3.2 (Chain group)**.** Let K be a simplicial complex. The *k-th chain group* $C_k(K)$ is the $\mathbb{F}$-vector space whose basis is the set of k-simplices of K. An element, a *k-chain*, is a formal sum

$$c = \sum_{\sigma} a_\sigma \, \sigma, \qquad a_\sigma \in \mathbb{F},$$

ranging over the k-simplices σ of K.

Remark 3.3. Over $\mathbb{F} = \mathbb{Z}_2$ a k-chain is determined by the subset of k-simplices it contains, and addition is componentwise modulo 2.

A chain group is thus a vector space whose basis vectors are the simplices themselves. Over $\mathbb{Z}_2$, a k-chain is essentially just a choice of which k-simplices to include, and adding two chains keeps only the simplices appearing in exactly one of the chains, since any simplex counted twice cancels when we work modulo 2. One may picture a k-chain as a highlighting of some subset of the k-dimensional pieces of the complex.

Definition 3.4 (Boundary map)**.** The *boundary map* $\partial_k : C_k(K) \to C_{k-1}(K)$ is the linear map defined on a k-simplex $\sigma = [v_0, \ldots, v_k]$ by

$$\partial_k \sigma = \sum_{i=0}^{k} (-1)^i \, [v_0, \ldots, \widehat{v_i}, \ldots, v_k],$$

where $\widehat{v_i}$ denotes omission of v_i. Over $\mathbb{F} = \mathbb{Z}_2$ the signs vanish and $\partial_k \sigma = \sum_{i=0}^{k} [v_0, \ldots, \widehat{v_i}, \ldots, v_k]$.

The boundary map sends each simplex to the sum of the faces that bound it: an edge maps to its two endpoints, a filled triangle to its three edges, and a tetrahedron to its four triangular faces. Extending linearly, the boundary of a chain is the sum of the boundaries of its pieces, with shared faces cancelling in pairs over $\mathbb{Z}_2$.

Example 3.5. Consider the filled triangle $\sigma = [a, b, c]$. Working over $\mathbb{Z}_2$, its boundary is

$$\partial_2 \sigma = [b, c] + [a, c] + [a, b],$$

the three edges forming its rim. Now consider the closed loop $c = [a, b] + [b, c] + [c, a]$ formed by these same three edges. Its boundary is

$$\partial_1 c = ([a] + [b]) + ([b] + [c]) + ([c] + [a]) = 2[a] + 2[b] + 2[c] = 0,$$

since every vertex appears twice. A chain with zero boundary, like this loop, is called a *cycle*.

Lemma 3.6 (Fundamental lemma of homology)**.** *For all k, $\partial_k \circ \partial_{k+1} = 0$. Consequently* $\operatorname{im} \partial_{k+1} \subseteq \ker \partial_k$.

Definition 3.7 (Cycles and boundaries)**.** The space of *k-cycles* is $Z_k(K) = \ker \partial_k$, and the space of *k-boundaries* is $B_k(K) = \operatorname{im} \partial_{k+1}$. By the fundamental lemma, $B_k(K) \subseteq Z_k(K)$.

Cycles are the closed chains that have no boundary of their own. Boundaries are the special cycles that arise as the rim of some higher-dimensional chain; they are the cycles that can be "filled in." Since the fundamental lemma guarantees $B_k \subseteq Z_k$, every fillable cycle is in particular a cycle.

Definition 3.8 (Homology group)**.** The *k-th homology group* of K over $\mathbb{F}$ is the quotient vector space

$$H_k(K) = Z_k(K)/B_k(K) = \ker \partial_k \,/\, \operatorname{im} \partial_{k+1}.$$

The homology group captures the k-dimensional features that are cycles but not boundaries, meaning they represent "holes" in dimension k.

Definition 3.9 (Betti number)**.** The *k-th Betti number* of K is

$$\beta_k = \dim_{\mathbb{F}} H_k(K) = \dim \ker \partial_k - \dim \operatorname{im} \partial_{k+1}.$$

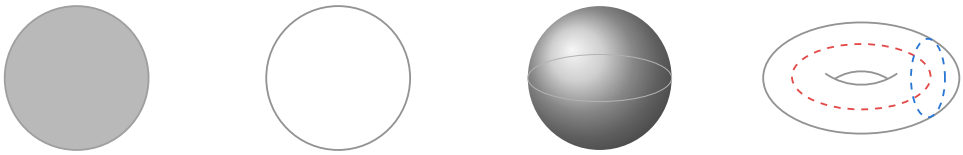
Intuitively, the k-th Betti number "counts" the independent k-dimensional holes. Specifically for the lower numbers, β_0 can be seen as the number of connected components, β_1 the number of independent loops, and β_2 the number of enclosed voids.

At this point it is fair to ask what these combinatorially defined groups have to do with topology. The chain groups were built from the simplices of K, and the same underlying space can be triangulated by many different complexes, so it is not obvious that we have computed an invariant of the space rather than of our chosen scaffolding. The resolution is the notion of *singular homology*. For an arbitrary topological space X, singular homology repeats the construction above with the k-simplices of K replaced by all continuous maps from the standard k-simplex into X; the resulting groups depend only on the topology of X and are invariant under continuous deformation (formally, *homotopy equivalence*) (Hatcher, 2002). The fundamental comparison theorem then states that for any simplicial complex K, the simplicial homology of K is isomorphic to the singular homology of its geometric realization $|K|$. Deep down, then, our algebra is topology: the combinatorial computation recovers an invariant of the underlying space that is unchanged under continuous deformation (in particular, under any homeomorphism), independent of the triangulation used to compute it. The following theorem records the consequence we rely on.

Theorem 3.10 (Homotopy invariance of homology)**.** *The homology groups of a simplicial complex depend only on the homotopy type of its geometric realization. That is, if the geometric realizations $|K|$ and $|L|$ can be continuously deformed into one another (they are* homotopy equivalent*), then $H_k(K) \cong H_k(L)$ for every k; in particular this holds whenever $|K|$ and $|L|$ are related by a continuous, continuously-invertible reshaping (they are* homeomorphic*)—for instance, a donut and a coffee mug.*

In other words, homotopy equivalent spaces, and homeomorphic spaces in particular, share the same homology and hence the same Betti numbers (Hatcher, 2002). This is what guarantees that the features we extract reflect the underlying space and not arbitrary modeling choices such as the triangulation. However, it is

important to understand that the converse fails: distinct spaces may share all of their Betti numbers, but not be homeomorphic.



$\beta_0 = 1$ $\beta_1 = 0$ $\beta_2 = 0$ | $\beta_0 = 1$ $\beta_1 = 1$ $\beta_2 = 0$ | $\beta_0 = 1$ $\beta_1 = 0$ $\beta_2 = 1$ | $\beta_0 = 1$ $\beta_1 = 2$ $\beta_2 = 1$

FIGURE 2. Betti numbers $\beta_0, \beta_1, \beta_2$ of four common spaces (left to right): a disk, a circle, a sphere, and a torus. The two dashed loops on the torus indicate its two independent 1-cycles.

Example 3.11. Figure 2 collects the Betti numbers of four familiar spaces. The disk can be crushed into a single point, so it forms a single connected component, contains no loop that fails to bound, and encloses no void, so $\beta_0 = 1$ and $\beta_1 = \beta_2 = 0$. The circle S^1 is likewise connected, but it carries one essential loop, giving $\beta_0 = 1$, $\beta_1 = 1$, and $\beta_2 = 0$. The sphere S^2 is connected and has no essential loops. Every loop on its surface contracts to a point, but it encloses a single two-dimensional void, so $\beta_0 = 1$, $\beta_1 = 0$, and $\beta_2 = 1$. Finally, the torus is connected and encloses one void, but it carries two independent loops: one passing through the central hole and one wrapping around the tube, neither deformable into the other. Thus, $\beta_0 = 1$, $\beta_1 = 2$, and $\beta_2 = 1$.

4. Point-Cloud Data

In practice, the data we wish to study rarely arrives as a smooth shape. It arrives as a point cloud, a finite collection of points scattered through a space such as $\mathbb{R}^d$. Each point is just a position given by its coordinates, and some metric that we choose lets us say how near or far any two of them are. On its own, a point cloud records only where the data lies, not how those locations are connected or what overall form they trace out. To recover the structure hidden in the data, we must supply that connectivity ourselves, by joining nearby points with edges, filling in a triangle wherever three points are mutually close, and continuing with higher-dimensional analogues. Fixing a scale, we declare nearby points to be connected and fill in simplices among mutually close points, so that the resulting complex captures the proximity structure of the cloud.

Definition 4.1 (Point cloud)**.** A *point cloud* is a finite subset $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$, regarded as a finite metric space with the inherited metric.

Definition 4.2 (Vietoris–Rips complex)**.** Let X be a point cloud and $\varepsilon \geq 0$ a scale parameter. The *Vietoris–Rips complex* $\mathrm{VR}_\varepsilon(X)$ is the abstract simplicial complex whose simplices are the finite subsets $\{x_{i_0}, \dots, x_{i_k}\} \subseteq X$ with

$$d(x_{i_j}, x_{i_\ell}) \leq \varepsilon \quad \text{for all } 0 \leq j, \ell \leq k.$$

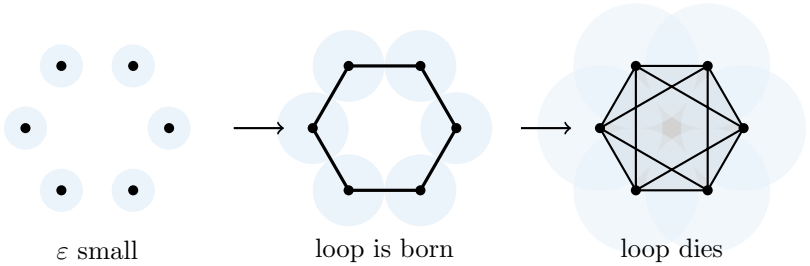


FIGURE 3. The Vietoris–Rips filtration of points sampled from a circle, as the scale ε grows (left to right). Each point carries a ball of radius $\varepsilon/2$, and an edge appears once two balls meet. The points are first isolated, then form a loop ($\beta_1 = 1$), and finally the loop fills in and dies ($\beta_1 = 0$).

Example 4.3. Let $X = \{A, B, C, D\}$ be the four corners of a unit square, with $A = (0, 0)$, $B = (1, 0)$, $C = (1, 1)$, and $D = (0, 1)$. The four sides have length 1 and the two diagonals have length $\sqrt{2}$. At scale $\varepsilon = 1$, the complex $\mathrm{VR}_1(X)$ contains the four vertices and the four side edges AB, BC, CD, DA, but no diagonal and no triangle, since any three corners include a diagonal of length $\sqrt{2} > 1$. Geometrically this is a hollow square, so a one-dimensional loop appears ($\beta_1 = 1$). At scale $\varepsilon = \sqrt{2}$, every pair of points lies within ε, so $\mathrm{VR}_{\sqrt{2}}(X)$ contains all edges, all triangles, and the full 3-simplex on $\{A, B, C, D\}$; the loop is filled in and disappears ($\beta_1 = 0$). The loop is therefore born at $\varepsilon = 1$ and dies at $\varepsilon = \sqrt{2}$, with persistence $\sqrt{2} - 1$.

Definition 4.4 (Čech complex)**.** Let X be a point cloud and $\varepsilon \geq 0$. The *Čech complex* $\check{\mathrm{C}}_\varepsilon(X)$ is the abstract simplicial complex whose simplices are the finite subsets $\{x_{i_0}, \ldots, x_{i_k}\} \subseteq X$ for which the closed balls of radius ε have a common point:

$$\bigcap_{j=0}^{k} \overline{B}(x_{i_j}, \varepsilon) \neq \emptyset.$$

Rather than the easy-to-understand Vietoris–Rips complex, the Čech complex instead fills in a simplex when the ε-balls around its vertices share a common point. By the Nerve Theorem (Borsuk, 1948; Edelsbrunner et al., 2002) it is homotopy equivalent to the union of those balls, so it captures the underlying topology more accurately than the Vietoris–Rips complex. The main difference between the two is that the Čech complex is more expensive to compute.

Definition 4.5 (Filtration)**.** A *filtration* of a simplicial complex K is a nested family of subcomplexes indexed by a parameter $\varepsilon \in \mathbb{R}$,

$$\emptyset = K_{\varepsilon_0} \subseteq K_{\varepsilon_1} \subseteq \cdots \subseteq K_{\varepsilon_m} = K, \qquad \varepsilon_0 \leq \varepsilon_1 \leq \cdots \leq \varepsilon_m.$$

For a point cloud X, the Vietoris–Rips filtration is $\{\mathrm{VR}_\varepsilon(X)\}_{\varepsilon \geq 0}$, with $\mathrm{VR}_\varepsilon(X) \subseteq \mathrm{VR}_{\varepsilon'}(X)$ whenever $\varepsilon \leq \varepsilon'$.

Essentially, what filtration accomplishes is giving us a way to track the lifetime of topological features. Instead of considering a single ε, we let it grow from 0. At small ε, our complex is simply a set of points. But at large ε, all of the points are connected to each other.

5. Persistent Homology

A single simplicial complex provides a snapshot of the data at one scale. However, using filtration we can find a whole family of them. Persistent homology tracks when each topological feature is born and when it dies, so that features which persist across a wide range of scales can be distinguished from short-lived ones attributable to noise.

Definition 5.1 (Persistence module)**.** Applying k-th homology with field coefficients to a filtration $\{K_{\varepsilon_i}\}$ yields a sequence of vector spaces and linear maps induced by the inclusions $K_{\varepsilon_i} \hookrightarrow K_{\varepsilon_{i+1}}$:

$$H_k(K_{\varepsilon_0}) \to H_k(K_{\varepsilon_1}) \to \cdots \to H_k(K_{\varepsilon_m}).$$

This is the *k-th persistent homology* of the filtration; abstractly, a *persistence module.*

Definition 5.2 (Birth, death, and persistence)**.** A homology class α is *born* at ε_b if it lies in $H_k(K_{\varepsilon_b})$ but not in the image of the map from $H_k(K_{\varepsilon_{b-1}})$. It *dies* at ε_d if it becomes trivial or merges with an older class in $H_k(K_{\varepsilon_d})$ but not before. Its *persistence* is $\varepsilon_d - \varepsilon_b$.

A feature is born at the scale where it first appears and dies at the scale where it is filled in, where it either becomes a boundary or is merged into an older feature. The difference between these two scales, its persistence, measures how robust the feature is. Long-persisting features reflect structure in the data, whereas features that are born and die almost immediately are generally noise. This is the backbone for why we can use persistent homology within financial market data.

Theorem 5.3 (Structure theorem for persistence modules, Zomorodian and Carlsson (Zomorodian and Carlsson, 2005))**.** *Let $\mathbb{F}$ be a field and let $\mathbb{V} : V_0 \to V_1 \to \cdots \to V_m$ be a persistence module of finite-dimensional $\mathbb{F}$-vector spaces. Then $\mathbb{V}$ decomposes as a direct sum of interval modules,*

$$\mathbb{V} \cong \bigoplus_j \mathbb{I}[b_j, d_j),$$

and this decomposition is unique up to reordering. Here $\mathbb{I}[b, d)$ equals $\mathbb{F}$ on indices $b \le i < d$, equals 0 elsewhere, and has identity maps between consecutive nonzero terms.

This theorem is what makes persistent homology well-defined and computable. It says that the entire persistence module breaks up uniquely into independent pieces, each of which is simply a feature that is born at some scale b_j and dies at some scale d_j. The whole of the persistent homology is therefore captured by the resulting collection of intervals $[b_j, d_j)$, which we now visualize.

Definition 5.4 (Persistence barcode)**.** The *persistence barcode* in degree k is the multiset of intervals $\{[b_j, d_j)\}_j$ from the interval decomposition of the degree-k persistence module, drawn as horizontal segments.

A barcode displays each feature as a horizontal bar stretching from its birth to its death. Long bars correspond to persistent, significant features; short bars correspond to noise. Counting the bars present at a given scale recovers the Betti numbers of the complex at that scale.

Definition 5.5 (Persistence diagram)**.** The *persistence diagram* $\mathrm{Dgm}_k(X)$ is the multiset

$$\mathrm{Dgm}_k(X) = \{(b_j, d_j)\}_j \subset \overline{\mathbb{R}}^2,$$

one point per interval $[b_j, d_j)$ of the degree-k barcode, together with every point of the diagonal $\Delta = \{(t,t) : t \in \mathbb{R}\}$ counted with infinite multiplicity. Each off-diagonal point represents a k-dimensional feature with birth b_j and death d_j.

A persistence diagram encodes the same information as a barcode, but plots each feature as the point (b_j, d_j) in the plane. Because death cannot precede birth, every point lies above the diagonal, and a point's distance above the diagonal is proportional to its persistence. Significant features therefore appear far from the diagonal, while noise clusters just above it.

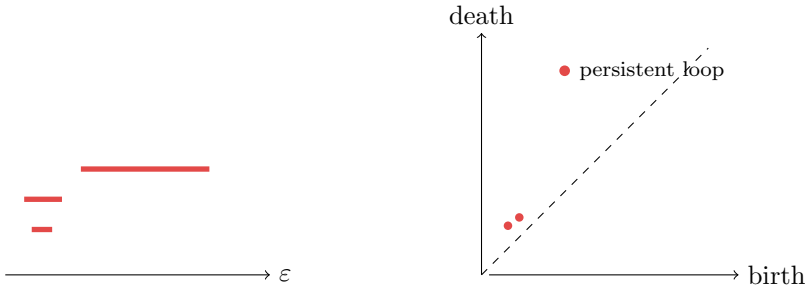


FIGURE 4. The same degree-1 persistent homology shown two ways. Left: a barcode, where each feature is a horizontal bar from birth to death. Right: the persistence diagram, where each feature is a point (birth, death) above the diagonal. The long bar and the point far from the diagonal are the persistent loop; the short bars and the points near the diagonal are noise.

6. Distances Between Persistence Diagrams

To actually be able to use persistence diagrams quantitatively, we need a way to measure the distance between two diagrams. Below, we introduce the primary methods to do so. Both distances are in fact defined for arbitrary finite multisets of points in the plane; nothing about them is special to persistence diagrams, which are simply the multisets we care about.

Definition 6.1 (Bottleneck distance)**.** Let D_1 and D_2 be two finite multisets of points in $\mathbb{R}^2$; in our applications these will be persistence diagrams. The *bottleneck distance* between them is

$$W_\infty(D_1, D_2) = \inf_{\gamma} \sup_{x \in D_1} \|x - \gamma(x)\|_\infty ,$$

the infimum over all bijections $\gamma : D_1 \cup \Delta \to D_2 \cup \Delta$, where $\|\cdot\|_\infty$ is the ℓ^∞ norm on $\mathbb{R}^2$.

The bottleneck distance pairs the points of the two diagrams, allowing any point to be matched instead to the diagonal, which represents a feature with no counterpart, and measures the largest distance any point must travel under the best such pairing. Matching a point to the diagonal costs only its distance to the diagonal, so low-persistence points, lying near the diagonal, contribute little; the bottleneck distance is governed by the single most significant feature that differs between the diagrams.

Definition 6.2 (p-Wasserstein distance)**.** Let D_1 and D_2 be two finite multisets of points in $\mathbb{R}^2$ and let $1 \le p < \infty$. The *p-Wasserstein distance* between them is

$$W_p(D_1, D_2) = \left(\inf_{\gamma} \sum_{x \in D_1} \|x - \gamma(x)\|_\infty^p \right)^{1/p},$$

the infimum over all bijections $\gamma : D_1 \cup \Delta \to D_2 \cup \Delta$.

The p-Wasserstein distance uses the same matching idea but sums the p-th powers of *all* the matched distances, rather than taking only the largest. It therefore measures the total reconfiguration of the diagram, accounting for every feature rather than just the most extreme one, and recovers the bottleneck distance in the limit $p \to \infty$. Where one wishes to detect aggregate change across many features at once, a finite p is the more informative choice.

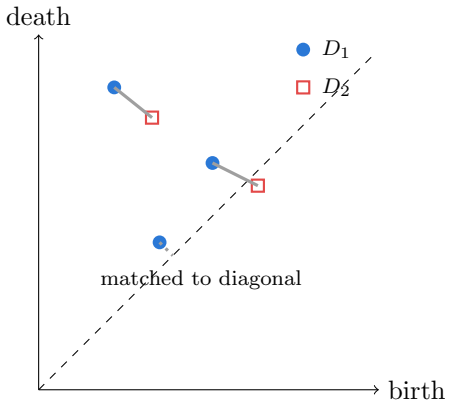


FIGURE 5. Computing a distance between two persistence diagrams D_1 (blue dots) and D_2 (red squares). Points are matched optimally, and a point with no good partner is matched to the diagonal at a cost equal to its distance from it. The bottleneck distance is the largest matched distance; the p-Wasserstein distance aggregates all of them.

Example 6.3. Let $D_1 = \{(0,6),(0,2)\}$ and $D_2 = \{(0,5)\}$, and recall that under the ℓ^∞ ground metric the distance from a point (b,d) to the diagonal is $(d-b)/2$. Compare the two possible matchings. Matching $(0,6)$ with $(0,5)$ costs $\|(0,6)-(0,5)\|_\infty = 1$, leaving $(0,2)$ to be sent to the diagonal at cost $(2-0)/2 = 1$; matching $(0,2)$ with $(0,5)$ instead costs 3 and sends $(0,6)$ to the diagonal at cost

3. The first matching is optimal, so

$$W_\infty(D_1, D_2) = \max\{1, 1\} = 1, \qquad W_1(D_1, D_2) = 1 + 1 = 2.$$

Notice how the diagonal absorbs the unmatched, low-persistence point cheaply, which is exactly what keeps these distances stable against noise.

Theorem 6.4 (Bottleneck stability, Cohen-Steiner, Edelsbrunner, and Harer (Cohen-Steiner et al., 2007))**.** *Let X be a triangulable topological space and let $f, g : X \to \mathbb{R}$ be tame functions. Then the persistence diagrams of their sublevel-set filtrations satisfy*

$$W_\infty\big(\mathrm{Dgm}(f), \mathrm{Dgm}(g)\big) \le \|f - g\|_\infty .$$

This stability theorem guarantees that the persistence diagram depends continuously (specifically 1-Lipschitz-continuously) on the input: a small perturbation of the data can move the diagram by only a correspondingly small amount, as shown by Cohen-Steiner, Edelsbrunner, and Harer (Cohen-Steiner et al., 2007). In particular, the significant, high-persistence features cannot be created or destroyed by noise, which is precisely why long bars can be trusted as actual structure.

Theorem 6.5 (Wasserstein stability, Cohen-Steiner, Edelsbrunner, Harer, and Mileyko (Cohen-Steiner et al., 2010))**.** *Let X be a triangulable, compact metric space with bounded degree-k total persistence for some $k \ge 1$, and let $f, g : X \to \mathbb{R}$ be two tame Lipschitz functions. Then for every $p \ge k$,*

$$W_p\big(\mathrm{Dgm}(f), \mathrm{Dgm}(g)\big) \le C^{1/p}\ \|f - g\|_\infty^{1-k/p},$$

where $C = C_X \max\{\mathrm{Lip}(f)^k, \mathrm{Lip}(g)^k\}$.

An analogous guarantee, due to Cohen-Steiner, Edelsbrunner, Harer, and Mileyko (Cohen-Steiner et al., 2010), holds for the Wasserstein distances under mild regularity, ensuring that aggregate measures of topological change are likewise robust to perturbation. This is what justifies using the Wasserstein distance between the diagrams of successive time windows as a stable signal of change in evolving data.

7. Vectorization of Persistence Diagrams

Persistence diagrams equipped with the distances above form a metric space, but not a vector space. There is no straightforward way to add or average diagrams, and most statistical and machine-learning methods require their inputs to be vectors. To make persistence usable as a feature in such methods, we need to vectorize the diagram, mapping it to an element of a vector space while retaining its topological content. We describe three standard constructions.

Definition 7.1 (Persistence landscape)**.** Let $D = \{(b_i, d_i)\}_{i=1}^m$ be a persistence diagram. For each point define the tent function

$$\Lambda_i(t) = \max\big\{0,\ \min(t - b_i,\ d_i - t)\big\}, \qquad t \in \mathbb{R}.$$

The *persistence landscape* of D is the sequence of functions $\lambda = (\lambda_\ell)_{\ell \ge 1}$, where $\lambda_\ell(t)$ is the ℓ-th largest value among $\Lambda_1(t), \dots, \Lambda_m(t)$ (and 0 if fewer than ℓ of them are nonzero).

The persistence landscape replaces each point of the diagram with a piecewise-linear tent function and then records, at each height, the successive maxima of these tents. The result is a sequence of functions lying in a vector space, so that landscapes may be added, averaged, and subjected to statistical analysis, with the most prominent features producing the tallest tents. The object that enters our models is not the landscape itself but its norm. Each λ_ℓ is compactly supported and piecewise linear, hence lies in $L^p(\mathbb{R})$, so the landscape has a well-defined L^p norm

$$\|\lambda\|_p = \left(\sum_{\ell \geq 1} \int_{\mathbb{R}} |\lambda_\ell(t)|^p \, dt \right)^{1/p},$$

which we evaluate at $p = 2$. Bubenik (Bubenik, 2015) shows this norm is stable under perturbations of the underlying diagram, so it inherits the robustness we demand of every feature in our pipeline.

Definition 7.2 (Persistence entropy)**.** Let $D = \{(b_i, d_i)\}_{i=1}^m$ be a persistence diagram with finite lifespans $\ell_i = d_i - b_i > 0$ (any essential class of infinite persistence is omitted), and let $L = \sum_{j=1}^m \ell_j$ denote the total persistence. The *persistence entropy* of D is the Shannon entropy of the normalized lifespans,

$$\mathrm{PE}(D) = -\sum_{i=1}^{m} p_i \log p_i, \qquad p_i = \frac{\ell_i}{L}.$$

Persistence entropy reduces a diagram to a single number recording how its total persistence is *distributed* across features, rather than how large that persistence is. It approaches 0 when one feature dominates and is maximized when many features share comparable lifespans, so it is naturally read as a measure of topological complexity. Introduced by Rucco et al. (Rucco et al., 2016), it sits on the same footing as the diagram distances of Section 6: Atienza, González-Díaz and Soriano-Trigueros (Atienza et al., 2020) establish bounds under which a small perturbation of the diagram produces only a small change in the entropy.

Definition 7.3 (Persistence image)**.** Map each point $(b, d) \in D$ to birth–persistence coordinates $(b,\ d-b)$. Fix a nonnegative weight function w and a bandwidth $\sigma > 0$, and form the surface

$$\rho(z) = \sum_{(b,d) \in D} w(b,\ d-b)\, g_\sigma\big(z - (b,\ d-b)\big), \qquad z \in \mathbb{R}^2,$$

where g_σ is a 2-dimensional Gaussian of bandwidth σ. Discretizing ρ over a fixed pixel grid yields the *persistence image*, a finite-dimensional vector.

The persistence image instead spreads each point of the diagram into a small Gaussian bump, weighted so that more persistent features count for more, and sums these into a smooth surface, following Adams et al. (Adams et al., 2017). Because every diagram is mapped to a vector of the same dimension, persistence images can be fed directly into standard machine-learning models.

8. Methodology

We will now describe the method used to implement the topological concepts described above in widely-used machine learning methods. Throughout, we are trying to analyze whether these augmented models will predict the realized volatility

of cryptocurrency better than the established methods. All of the features we construct are evaluated at time t using only information available up to and including t.

8.1. **Data and cleaning.** Our data consists of one-minute open-high-low-close bars for BTC/USDT and ETH/USDT pulled from Binance US over 2020-01-01 to 2023-12-31, a span deliberately chosen to contain the May 2021 sell-off, the Terra/Luna collapse of May 2022, and the FTX collapse of November 2022. Binance US is materially thinner than the primary Binance venue (used here only because the latter is geo-blocked in the United States), which introduces two artifacts that the realized-volatility estimator below is sensitive to. First, isolated erroneous prints in the high or low column are invisible to a close-to-close filter but enter the range-based estimator directly. We thus repair any wick exceeding the candle body by more than 15%, clipping it back to the body. The wick repair alters 5 of 2,100,035 BTC bars and 13 of 2,102,659 ETH bars. Second, a large fraction of one-minute bars carry no trade ($O = H = L = C$), which drives the estimator to zero and produces $\log 0$ artifacts. Due to this, we create a variance floor of 10^{-8} (in five-minute variance units, roughly 3% in annualized volatility terms) applied to the target before we take any logarithms.

8.2. **Target variable.** To sit outside the microstructure-noise regime we subsample the one-minute bars to a five-minute grid before estimating volatility. For a five-minute bar with prices (O_j, H_j, L_j, C_j) define the Rogers–Satchell range term

$$\mathrm{RS}_j \;=\; \ln\tfrac{H_j}{O_j}\,\ln\tfrac{H_j}{C_j} \;+\; \ln\tfrac{L_j}{O_j}\,\ln\tfrac{L_j}{C_j},$$

which is unbiased for the intrabar variance and exploits the full bar rather than the close alone (Rogers and Satchell, 1991). Over a window of n five-minute bars we form the Yang–Zhang-type estimator

$$\widehat{\mathrm{RV}}_n \;=\; k\,\mathrm{Var}\big(\ln\tfrac{C_j}{O_j}\big) \;+\; (1-k)\,\overline{\mathrm{RS}}, \qquad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}},$$

where the close-to-open (overnight) component of the classical estimator vanishes because the asset trades continuously (Yang and Zhang, 2000). The target is the forward realized volatility over the next thirty minutes, i.e. $\widehat{\mathrm{RV}}$ computed on the six bars $(t,\, t + 30\text{ min}]$. We forecast $\log \mathrm{RV}$, which is closer to Gaussian and standard in this literature (Andersen et al., 2003). We want to make sure that the units are well understood. $\widehat{\mathrm{RV}}$ is a variance-type quantity, and every model, loss, and table in this paper operates on this variance scale. Therefore, $\log \mathrm{RV}$ always denotes log-variance. Where a figure displays a volatility for visual readability, it plots the square root $\sqrt{\widehat{\mathrm{RV}}}$ expressed in percent, and its caption states the transform. Since $\ln\sqrt{v} = \frac{1}{2}\ln v$, the two scales are related by a constant factor in log space, so forecast rankings are unaffected by the choice.

8.3. **Benchmark models.** Our first benchmark is Corsi's Heterogeneous Autoregressive model of realized volatility (Corsi, 2009), an ordinary least-squares regression

$$\log \mathrm{RV}_{t+h} = c + \beta_S\,\log \mathrm{RV}_t^{(S)} + \beta_M\,\log \mathrm{RV}_t^{(M)} + \beta_L\,\log \mathrm{RV}_t^{(L)} + \varepsilon_t,$$

where $\mathrm{RV}^{(S)}, \mathrm{RV}^{(M)}, \mathrm{RV}^{(L)}$ are backward-looking realized volatilities over the last 5, 60, and 480 minutes, capturing the long memory of volatility across time scales.

Its linearity makes it a demanding benchmark that is notoriously hard to beat out of sample. Our second benchmark is Chen and Guestrin's gradient-boosted regression-tree model, XGBoost (Chen and Guestrin, 2016), fitted to the same three inputs. Its hyperparameters are relatively conservative and fixed: 300 trees of maximum depth 4, learning rate 0.05, row and column subsampling of 0.8, minimum child weight 10, and L^2 regularization $\lambda = 1$. These values were never changed throughout testing, so any difference between a baseline and its augmentation can be attributed to the features added. This model supplies the nonlinear counterpart to HAR in which, following Souto (Souto, 2023), topological features are expected to contribute most. Both models predict log RV and are exponentiated to return volatility on its natural scale, so that the evaluation below treats them identically.

8.4. **Topological feature construction.** We can get from a scalar return series to a point cloud through Takens' time-delay embedding (Takens, 1981). Fix an emission time t on the five-minute grid. Let (r_i) be the one-minute log-returns in the window $(t - 60 \text{ min}, t]$, each clipped in magnitude at 20% to remove residual bad prints and rescaled by a common constant so that returns are expressed in percent. This clipping actually never binds at all (the largest absolute one-minute return in the sample is 6.9% for BTC and 8.5% for ETH). Any clipping threshold in a wide band therefore yields identical data. With embedding dimension $m = 3$ and delay $\tau = 1$ we form the delay vectors

$$x_i \;=\; \big(r_i,\, r_{i-\tau},\, \ldots,\, r_{i-(m-1)\tau}\big) \in \mathbb{R}^m,$$

whose collection $X_t = \{x_i\}$ is the point cloud summarizing the local return dynamics of the window. On X_t we build the Vietoris–Rips filtration, computed via Bauer's Ripser (Bauer, 2021), over $\mathbb{Z}_2$ and compute persistent homology in degrees zero and one, yielding the persistence diagrams $D_0(t)$ and $D_1(t)$.

From each window we extract a twelve-dimensional feature vector. Writing $\ell_i = d_i - b_i$ for the lifespan of a point (b_i, d_i) and omitting the essential class of infinite persistence, we record, in each degree $k \in \{0, 1\}$ where applicable:

- the *total persistence* $\sum_i \ell_i$ and, in degree one, the *maximum persistence* $\max_i \ell_i$;
- the *significant-loop count* $|\{i : \ell_i > \vartheta\}|$ in degree one, with $\vartheta = 0.10$ fixed a priori in the units of the scaled returns (returns are multiplied by 100, so ϑ corresponds to a lifespan of ten basis points in raw-return terms); the threshold discards features indistinguishable from noise (cf. the Betti number);
- the *persistence entropy* $\mathrm{PE}(D_k)$;
- the *persistence-landscape* L^2 norm $\|\lambda\|_2$.

We also record the 2-Wasserstein distances $W_2\big(D_0(t), D_0(t-1)\big)$ and $W_2\big(D_1(t), D_1(t-1)\big)$ and the bottleneck distance $W_\infty\big(D_1(t), D_1(t-1)\big)$. By the stability theorems in Section 6, these are stable functions of the data, so a large value reflects an actual signal rather than sampling noise (this is the topological analogue of a regime-change signal).

Because of our construction, a window with fewer than eight distinct return values (a flat, illiquid stretch) yields a degenerate point cloud. Thus, we flag such windows with a binary indicator, emit zeros for their topological features, and compute the change distances against the most recent non-degenerate diagram, so

that liquidity gaps do not masquerade as topological change. Because the sixty-minute windows aggregate many bars, degeneracy is rarer than the raw no-trade rates (37% of BTC and 55% of ETH one-minute bars). Therefore, only 0.8% of BTC and 6.8% of ETH windows are flagged. Section 9.5 reports an ablation showing the indicator contributes nothing to forecast accuracy, so we can confirm that results do not rest on this liquidity information.

The resulting feature vector is indexed on the same five-minute grid as the target and joined to it, so that baseline and topology-augmented models are evaluated on identical observations.

TABLE 1. The twelve topological features extracted from each window. Lifespans $\ell_i = d_i - b_i$ are taken over the finite points of the indicated diagram; $\vartheta = 0.10$. Landscape norms integrate the first five landscape layers on a uniform grid over the diagram's support. Features marked † form the four-feature subset supplied to the linear model.

Feature	Degree	Definition
Wasserstein change†	1	$W_2\big(D_1(t),\, D_1(t-1)\big)$
Wasserstein change	0	$W_2\big(D_0(t),\, D_0(t-1)\big)$
Bottleneck change	1	$W_\infty\big(D_1(t),\, D_1(t-1)\big)$
Persistence entropy	0	$\mathrm{PE}(D_0(t))$
Persistence entropy†	1	$\mathrm{PE}(D_1(t))$
Maximum persistence†	1	$\max_i \ell_i$
Total persistence	0	$\sum_i \ell_i$
Total persistence	1	$\sum_i \ell_i$
Significant-loop count†	1	$\lvert\{i : \ell_i > \vartheta\}\rvert$
Landscape L^2 norm	0	$\lVert\lambda(D_0(t))\rVert_2$
Landscape L^2 norm	1	$\lVert\lambda(D_1(t))\rVert_2$
Degeneracy indicator	–	$\mathbf{1}\{\text{window flat}\}$

Since these methods would in practice be applied in a high-frequency financial setting, we briefly mention their computational cost. One window's full pipeline takes under 3 milliseconds on a single consumer CPU core, and the entire four-year history of one asset rebuilds in about thirteen minutes on eight cores, so the feature construction is computationally inexpensive.

8.5. **Controls to account for frequency introduced by topological features.** Before we make any topological claims, we have to address an important hazard. The degree-zero total persistence, which measures the dispersion of the point cloud, correlates at 0.83 (BTC) and 0.70 (ETH) with the logarithm of one-minute realized variance over the trailing hour, which the five-minute baseline models never receive. Thus, any improvement in our models could be due to the topology smuggling in finer-frequency volatility. To separate the two we introduce non-topological controls computed from the same one-minute returns over trailing windows of 15 and 60 minutes: the ordinary realized variance $\sum_i r_i^2$; the jump-robust bipower variation of Barndorff-Nielsen and Shephard (Barndorff-Nielsen and Shephard, 2004)

$$\mathrm{BV}_W \;=\; \frac{\pi}{2} \sum_i |r_i|\,|r_{i-1}|;$$

and the range-based realized range of Christensen and Podolskij (Christensen and Podolskij, 2007) computed on repaired high/low prices,

$$\mathrm{RR}_W \;=\; \frac{1}{4\ln 2}\sum_i \big(\ln H_i - \ln L_i\big)^2.$$

The latter two are especially sturdy against the single freak prints that a dispersion statistic might exploit, and thus form the best benchmarks. By supplying these controls to both sides of a comparison, we isolate specifically the contribution of topology.

8.6. **Model configurations and forecast evaluation.** For each of our two baselines (HAR-RV, XGBoost), we score the accuracy of the three realized-volatility lags alone, then with the two plain one-minute controls added (RV1), then with the full set of six controls (RVrob), and finally with the topological features added. Because the linear model is prone to collinearity, its topological augmentation uses a particular subset of our features marked in Table 1, whereas the gradient-boosted model, untroubled by correlated inputs, receives all twelve. We selected these four in order to span the distinct kinds of topological summary we compute. To confirm this choice, we computed variance inflation factors (VIFs) on the training span. The four selected had VIFs between 2.0 and 4.3 on both assets, whereas entering all twelve raises VIFs within the topological block to 19 (BTC) and 51 (ETH).

All models are evaluated by expanding-window walk-forward out of sample. At each step the model is fitted on all data preceding the test window and used to predict the next thirty days, after which the window advances. The model is refitted every thirty days, with the out-of-sample period beginning 2022-01-01. Because the target looks forward six bars, the last observations of each training fold carry labels that overlap the test window; we therefore purge those bars from training, so that no training label peeks into the evaluation period, following López de Prado (López de Prado, 2018). Forecasts are scored by two losses: the quasi-likelihood

$$\mathrm{QLIKE} \;=\; \frac{1}{N}\sum_t \left(\frac{\mathrm{RV}_t}{\widehat{\mathrm{RV}}_t} - \ln\frac{\mathrm{RV}_t}{\widehat{\mathrm{RV}}_t} - 1\right),$$

where RV_t and $\widehat{\mathrm{RV}}_t$ are on the variance scale of Section 8.2. QLIKE is the proper and standard loss for volatility and penalizes under-prediction of variance asymmetrically. The second loss is the mean squared error of $\log \mathrm{RV}$, computed on log-variance, a symmetric complement. We additionally report both losses restricted to the turbulent sub-periods named above.

To determine whether a difference in forecast accuracy is real rather than an artifact of the heavy serial correlation of intraday losses, we use the Diebold–Mariano test (Diebold and Mariano, 1995). For two models A and B let d_t be the difference of their per-observation quasi-likelihood losses; the statistic is

$$\mathrm{DM} \;=\; \frac{\bar d}{\sqrt{\widehat{\mathrm{LRV}}/N}}, \qquad \widehat{\mathrm{LRV}} \;=\; \gamma_0 + 2\sum_{k=1}^{h-1}\gamma_k,$$

where γ_k are the autocovariances of d_t and the truncation at $h-1$ reflects the moving-average dependence induced by our overlapping six-bar forecasts. We apply the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997) and refer the corrected statistic to Student's t distribution with $N-1$ degrees of freedom.

9. Results

9.1. **Initial evidence.** Before any forecasting, Figure 6 plots the degree-one topological-change signal $W_2\big(D_1(t), D_1(t-1)\big)$ against the realized volatility over the subsequent thirty minutes, across the full four-year sample for BTC/USDT. The two series co-move visibly (correlation +0.60 with log-RV), and the change signal spikes together with volatility during each of the three labelled stress episodes.

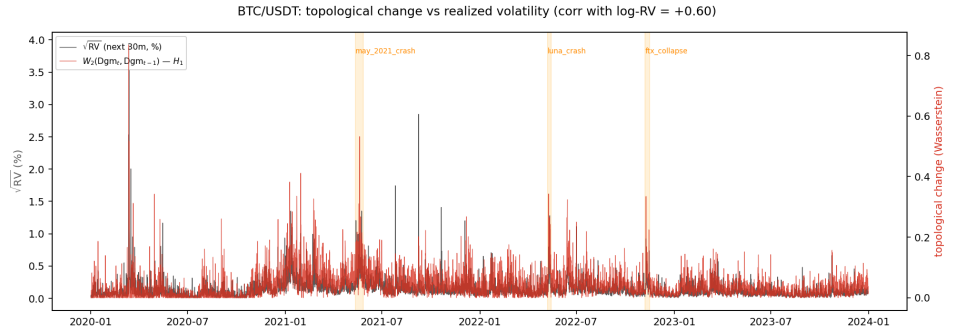


Figure 6. The Wasserstein topological-change signal (red) against the square root of the thirty-minute-ahead realized variance, expressed in percent (grey), BTC/USDT, 2020–2023. Shaded bands mark the May 2021 sell-off, the Terra/Luna collapse, and the FTX collapse.

9.2. **Out-of-sample forecast accuracy.** Table 2 reports QLIKE and MSE-log for all twelve model configurations over the two-year out-of-sample period (209,910 five-minute observations for BTC/USDT, 209,999 for ETH/USDT). Adding the twelve topological features to the plain baselines lowers QLIKE by 6.1%[1] (HAR-RV) and 6.9% (XGBoost) for BTC/USDT, and by 7.2% and 14.1% respectively for ETH/USDT. Consistent with Souto (Souto, 2023), the nonlinear model benefits roughly twice as much as the linear one.

Giving the baselines the robust one-minute controls of Section 8.5, with no topology at all, recovers a large share of the improvement: XGBoost+RVrob lowers QLIKE by 7.4% (BTC) and 10.0% (ETH) relative to plain XGBoost. Interestingly, the plain one-minute realized variance alone does not achieve this (for BTC it actually significantly worsens both baselines). Had we compared only the naive pair, we would have attributed to topology what is partly the work of a better volatility estimator.

However, topology does still survive the strongest control. Adding the topological features to the fully controlled models produces a further QLIKE reduction of 3.30% and 3.25% for the BTC models, 3.9% for the ETH linear model, and 5.9% for the ETH nonlinear model. The fully augmented nonlinear model is the best configuration in every column of Table 2, with an overall QLIKE 8.4% (BTC)

[1]Percentage changes are computed from unrounded losses. Reconstructing them from the four-decimal entries of Table 2 can differ by up to 0.1 percentage points.

and 12.2% (ETH) below the classical HAR-RV benchmark. Figure 7 displays this ladder.

TABLE 2. Out-of-sample forecast accuracy, 2022-01-01 to 2023-12-31. QLIKE and mean squared error on log RV; lower is better. RV1 denotes the plain one-minute realized-variance controls; RVrob the full robust control set (plain, bipower, range); TDA the topological features. Bold marks the best value per column.

	BTC/USDT		ETH/USDT	
Model	QLIKE	MSE-log	QLIKE	MSE-log
HAR-RV	0.6388	0.7250	1.0350	1.1065
HAR+TDA	0.6000	0.7298	0.9608	1.1127
HAR+RV1	0.6483	0.7218	1.0292	1.1082
HAR+RV1+TDA	0.6116	0.7214	0.9599	1.1111
HAR+RVrob	0.6273	0.7185	0.9750	1.1067
HAR+RVrob+TDA	0.6066	0.7207	0.9368	1.1082
XGBoost	0.6528	0.7220	1.0733	1.1157
XGBoost+TDA	0.6078	0.7092	0.9223	1.0806
XGBoost+RV1	0.6614	0.7165	1.0360	1.1028
XGBoost+RV1+TDA	0.6087	0.7045	0.9259	1.0732
XGBoost+RVrob	0.6045	0.6999	0.9656	1.0787
XGBoost+RVrob+TDA	**0.5849**	**0.6956**	**0.9090**	**1.0647**

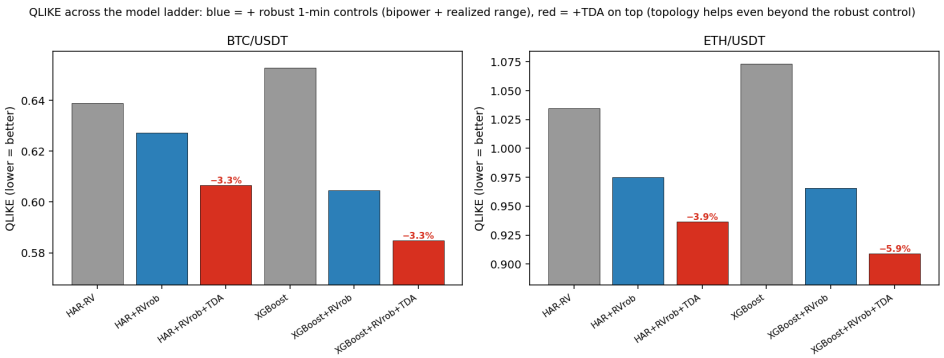


FIGURE 7. QLIKE across the model ladder. Grey: baseline; blue: with the robust one-minute controls; red: with the topological features added on top. The percentage above each red bar is the reduction attributable to topology alone.

9.3. **Statistical significance.** Table 3 reports the corrected Diebold–Mariano statistics of Section 8.6. Across the eight comparisons that add topological features (the first, second, fifth, and sixth rows of each panel), every augmentation is significant, with statistics between 7.6 and 19.2. The most important rows are the last

two for each asset, as they contain the robust control on both sides. We can see that they are clearly significant, with statistics of 9.85 and 8.94 (BTC) and 7.57 and 7.84 (ETH). The table also shows that for BTC/USDT, the plain one-minute control alone actually worsens both baselines (statistics -18.82 and -6.41), so the topological features are not merely proxying raw one-minute variance.

TABLE 3. Diebold–Mariano tests on the per-observation QLIKE differential (Harvey–Leybourne–Newbold corrected, HAC truncation at the forecast overlap). Positive Δ means the second model is better. All p-values are below 2×10^{-10}.

Asset	Comparison	Δ QLIKE	DM stat
BTC/USDT	HAR-RV → HAR+TDA	+0.0388	+14.31
BTC/USDT	XGBoost → XGBoost+TDA	+0.0450	+12.84
BTC/USDT	HAR-RV → HAR+RV1	−0.0095	−18.82
BTC/USDT	XGBoost → XGBoost+RV1	−0.0086	−6.41
BTC/USDT	HAR+RVrob → HAR+RVrob+TDA	+0.0207	+9.85
BTC/USDT	XGBoost+RVrob → XGBoost+RVrob+TDA	+0.0197	+8.94
ETH/USDT	HAR-RV → HAR+TDA	+0.0742	+10.69
ETH/USDT	XGBoost → XGBoost+TDA	+0.1510	+19.20
ETH/USDT	HAR-RV → HAR+RV1	+0.0057	+7.49
ETH/USDT	XGBoost → XGBoost+RV1	+0.0374	+14.35
ETH/USDT	HAR+RVrob → HAR+RVrob+TDA	+0.0381	+7.57
ETH/USDT	XGBoost+RVrob → XGBoost+RVrob+TDA	+0.0565	+7.84

9.4. **Where and when the improvements happen.** The improvement is concentrated where prior literature has found it. Holding the robust controls fixed, the topological features reduce the nonlinear model's QLIKE by 3.25% (BTC) and 5.9% (ETH) but its MSE-log by only 0.6% and 1.3%. For the linear models, MSE-log is essentially unchanged (moving slightly against the augmentation) even as QLIKE decreases by 3.30% and 3.9%. QLIKE is dominated by under-predicted volatility spikes while MSE-log is symmetric, so we conclude that topology helps chiefly in predicting large jumps, rather than small shifts in calm periods.

Figure 8 shows this explicitly by displaying the cumulative QLIKE gain of the topologically augmented nonlinear model over its controlled baseline across the out-of-sample period. The accuracy gain accumulates steadily over the two years, with visible accelerations during the Terra/Luna collapse, the June 2022 deleveraging, and the FTX collapse, and a flatter slope through the calmer conditions of 2023.

Table 4 restricts both losses to the two turbulent regimes that fall inside the out-of-sample period (the May 2021 episode predates it). The fully augmented nonlinear model attains the best QLIKE in all four regime cells, with topology contributing reductions over its controlled baseline of 5.2% (BTC) and 4.4% (ETH) during the Terra/Luna collapse and 0.6% and 2.9% during the FTX collapse. However, the topological augmentation of the linear model becomes worse during the most violent regime (ETH/USDT during FTX, 1.282 → 1.313), even while helping on average. Therefore, the gains from topology in extreme conditions are present mainly for the nonlinear model.

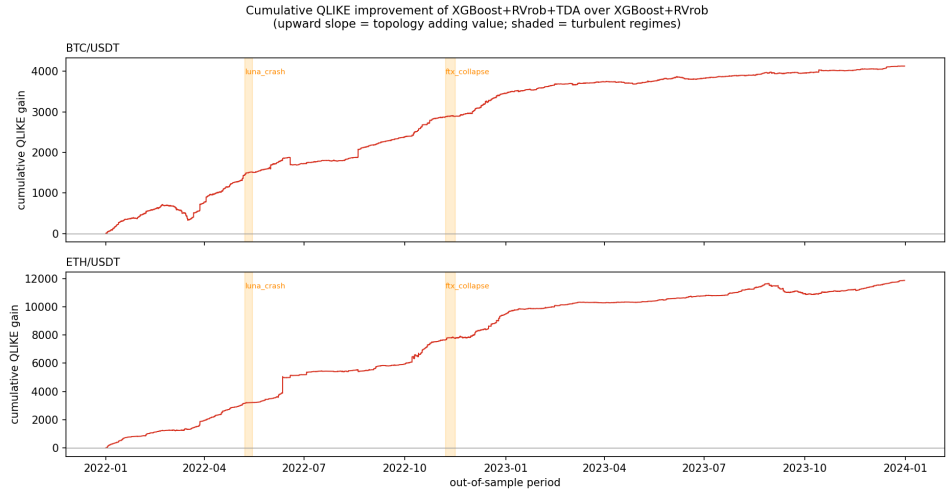


FIGURE 8. Cumulative QLIKE improvement of XGBoost+RVrob+TDA over XGBoost+RVrob. An upward slope indicates topology adding value; shaded bands are the labelled turbulent regimes.

TABLE 4. QLIKE within the turbulent out-of-sample regimes (Terra/Luna: 2022-05-08 to 2022-05-15, $n \approx 2{,}017$; FTX: 2022-11-07 to 2022-11-16, $n \approx 2{,}593$). Bold marks the best value per column.

	BTC/USDT		ETH/USDT	
Model	Luna	FTX	Luna	FTX
HAR-RV	0.5490	0.5884	0.5975	1.2618
HAR+RVrob	0.5401	0.5947	0.6017	1.2816
HAR+RVrob+TDA	0.5158	0.5960	0.5716	1.3127
XGBoost	0.5196	0.5696	0.5632	1.2612
XGBoost+RVrob	0.5025	0.5477	0.5492	1.2199
XGBoost+RVrob+TDA	**0.4765**	**0.5446**	**0.5250**	**1.1846**

It is worth mentioning that a venue-specific structural break falls inside the out-of-sample window. Following the SEC action against Binance and Binance US filed on 2023-06-05, liquidity on our venue deteriorated sharply. Median daily volume in the out-of-sample window fell by 82% for BTC/USDT and 90% for ETH/USDT relative to the pre-action period, the share of flat one-minute ETH bars rose from 44% to 70%, and the fraction of degenerate embedding windows rose from 0.15% to 2.6%. However, the largest topological gain occurs before the break. Splitting the controlled comparison at that date, the topological gain before the break is 3.9% for BTC and 7.5% for ETH, with Diebold–Mariano statistics of 8.5 and 7.3, while the gain after the break is 1.0% and 2.1%, with statistics of 3.1 each. This is also

why our sample ends at the close of 2023: post-2023 data on this venue is of lower quality.

9.5. **Which topological features carry the signal.** Figure 9 shows the individual feature importance of the fully controlled nonlinear model throughout training. We can first see that even with every volatility estimator available, topological features are still some of the most valuable features. For BTC/USDT the degree-zero and degree-one total persistence rank directly behind the volatility controls, and for ETH/USDT the degree-zero persistence entropy is the second most important feature overall. Persistence entropy is scale-free by construction (it depends only on the distribution of lifespans, not their magnitude), so its prominence cannot be explained by any volatility-proxy argument. We can also see that the degeneracy indicator ranks highly for ETH/USDT, suggesting that on a thin venue the liquidity structure itself is forecast-relevant.

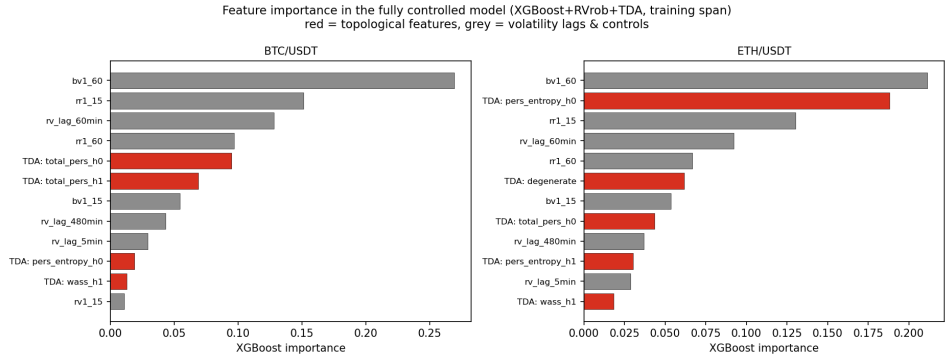


FIGURE 9. Feature importance in the fully controlled nonlinear model (XGBoost+RVrob+TDA), fitted on the training span. Red bars are topological features, blue bars are the one-minute volatility controls, and grey bars are the HAR lags.

We now must defend against some natural objections. First, because the degeneracy indicator is a liquidity flag rather than a topological quantity, we reran the fully controlled nonlinear model with the flag excluded. The gain over XGBoost+RVrob was unchanged (Diebold–Mariano statistics of 8.74 and 7.59 for BTC and ETH), and the flag's contribution, tested with the same Harvey–Leybourne–Newbold-corrected Diebold–Mariano procedure applied to the per-observation QLIKE differential between the with-flag and without-flag models, is indistinguishable from zero ($p = 0.63$ and $p = 0.59$). Its prominence in Figure 9 therefore reflects importance shared among correlated features rather than unique information. Second, to separate model class from feature count, we reran the fully controlled tree restricted to the same four topological features the linear model receives. It still beats its control decisively (statistics of 7.91 and 7.54), so the result does not depend on the wider feature set. However, its gain (2.2% and 3.1%) is no larger than the linear model's with the same features (3.3% and 3.9%). This is interesting, as the nonlinear model's advantage in the full specification therefore reflects its capacity to absorb the remaining eight collinear features rather than nonlinearity applied to the same inputs.

9.6. **Practical application and economic significance.** With roughly 210,000 observations per asset, DM statistics of the magnitude we report are to be expected even for economically small effects, since large samples shrink the standard error of the loss differential. The statistics show that there is some improvement, but not that the improvement matters. To address this, we transform the forecasts into two decision-relevant exercises, built on non-overlapping thirty-minute periods ($n \approx 35{,}000$ per asset).

First, we do a Value-at-Risk comparison, where each model's VaR at level α is z_α times its forecast volatility, using a normal quantile. Fat tails make every model under-cover, but the distributional assumption is shared, so the following ranking is informative. The fully augmented model is closest to nominal coverage in every case (violation rates of 3.50% and 7.22% for BTC at the 1% and 5% levels, against 3.56% and 7.37% for its control; 6.11% and 11.22% for ETH against 6.45% and 11.60%).

Second, we create a volatility-targeting exercise in the style of Fleming, Kirby, and Ostdiek (Fleming et al., 2001). A long-only strategy scales its position to a 40% annualized volatility target using each model's forecast, with leverage capped at five. The augmented model tracks the target most closely on both assets (56.1% realized against 56.8% for its control on BTC; 73.4% against 76.0% on ETH). All models overshoot because the range-based estimator understates close-to-close variance on a thin venue, an error common to every strategy compared. Under quadratic utility, the certainty-equivalent[2] gain from switching to the augmented model's forecasts is between 6 and 9 percent per year for BTC and between 17 and 32 percent per year for ETH across risk aversions $\gamma \in \{2, 6, 10\}$, gross of transaction costs. These figures are a unit translation of forecast quality rather than a claim of implementable alpha, but they indicate that the QLIKE improvements are of an economically significant magnitude.

10. Discussion

10.1. **Interpretation.** A very important caveat must be stated before discussing our results. Table 2 shows that much of the gain from simply adding topological features to our models can be attributed to the finer-frequency information that the models acquire along with those features. This is clearest when comparing the decrease in QLIKE from adding TDA to the plain model, to the model with RV1, and finally to the model with RVrob. Topological summaries of finer-frequency data inevitably carry information about that finer frequency, so any study that augments a coarse benchmark with such features must control for the frequency channel before attributing the gain to topology itself. Studies that omit this control will systematically overstate the value of topological methods.

However, with the control in place, we still find a statistically significant gain with TDA. Even with the most powerful control, adding the topological features still reduces QLIKE by 3.2% to 5.9%, with Diebold–Mariano statistics between 7.6

[2]At each period the strategy holds $w_t = \min\{5,\ \sigma^*/\hat{\sigma}_t\}$, where $\hat{\sigma}_t$ is the model's forecast volatility over the next thirty minutes and $\sigma^* = 0.40/\sqrt{17{,}520}$ is the per-period equivalent of a 40% annualized target (17,520 non-overlapping thirty-minute periods per year). Writing μ and σ for the annualized mean and standard deviation of the realized strategy return $w_t r_{t,t+30}$, quadratic utility gives the certainty equivalent $\mathrm{CE} = \mu - \frac{1}{2}\gamma\sigma^2$, and the reported figure is $\mathrm{CE}_{\mathrm{TDA}} - \mathrm{CE}_{\mathrm{control}}$.

and 9.9 (Table 3). Two properties of this residual argue that it reflects our point-cloud topology rather than merely a residual volatility proxy. First, in Figure 9, the degree-zero persistence entropy ranks second among all features for ETH/USDT despite depending only on the distribution of lifespans and not on their magnitude. Second, most of our gains appear in QLIKE rather than MSE-log. Since QLIKE penalizes under-predicted volatility spikes more heavily than the symmetric MSE-log, this asymmetry is difficult to reconcile with a pure volatility proxy, which would be expected to improve both losses.

Our results clarify the picture in a way that echoes Souto's findings (Souto, 2023). As Souto found for lower-frequency predictors, the benefit of topology in extreme conditions clearly belongs to the nonlinear model: out of all of our different models, the fully augmented XGBoost+RVrob+TDA attains the best QLIKE in every turbulent cell of Table 4, while its linear counterpart actually degrades during the same periods. If topological features encode early geometric signatures of large moves, it is perhaps unsurprising that a linear functional of them is too rigid to exploit that information when it matters most.

10.2. **Limitations.** The first limitation arises because the primary Binance venue is geo-blocked in the United States, and Binance US is much shallower: 37 percent of BTC one-minute bars and 55 percent of ETH bars contain no trade at all. A vendor-sourced dataset drawing on Binance's primary liquidity (via Kaiko, Tardis.dev, or similar) is a feasible alternative that would likely reduce the no-trade rate substantially. Our choice of Binance US reflects free, direct API access rather than a hard data-availability constraint. With cleaner data, we could find either that our topological gain shrinks (if part of the signal is a liquidity artifact) or that it increases (if cleaner clouds create sharper geometry). Additionally, we used default embedding parameters ($m = 3$, $\tau = 1$) rather than selecting them by mutual-information or false-nearest-neighbour criteria, and we fixed the significant-loop threshold rather than tuning it. Our two models also receive different collections of topological features, so the comparison between our linear and nonlinear findings is indicative rather than exact. Finally, we study two assets in a single asset class, albeit the two largest.

10.3. **Future work.** The persistence image of Definition 7.3 is a natural high-dimensional input for the gradient-boosted model and remains unused by our forecasts. Thus, in future work one could exploit this as a feature and see if it boosts the gains we found. Cryptocurrency options markets publish implied volatility for both of our assets. Since IV embeds the market's own forward-looking expectation, one could test whether TDA adds value beyond an implied-volatility control. Another extension would be to create a new point cloud with axes over several assets, which would bring the analysis closer to the topological tail dependence that motivated it, and to degree-two homology, whose computational cost we avoided here. A final direction would be to replicate the experiment on deeper venues and other asset classes, with embedding parameters selected by mutual information and false-nearest-neighbour criteria rather than fixed a priori.

11. Conclusion

This paper discusses the relatively novel integration of TDA into cryptocurrency market analysis, specifically regarding realized volatility predictions. The primary

contribution of this paper is the addition of topological features to existing machine learning models, whose aim is to predict realized volatility over short horizons. Specifically, we constructed twelve persistent homology summaries of our data, whose use as a genuine signal is supported by the stability theorems of Section 6.

While carrying out this test, we identified a methodological hazard that, to our knowledge, prior work in this field has not addressed. Persistent homology features calculated on a finer frequency than that given to the models necessarily hold information about that finer frequency, so a simple comparison can conflate the gain accrued from topology with the gain accrued from having finer data. In response, we created multiple controls, to fairly test solely the increase in accuracy gained from topological features.

Our empirical results show that with naive comparisons, persistent homology reduces QLIKE by 6 to 14 percent, with the nonlinear XGBoost model benefiting roughly twice as much as its linear HAR counterpart. However, approximately half of this benefit is reproduced with our set of robust controls, with the remaining 3.2 to 5.9 percent QLIKE reduction surviving our strongest control, with Diebold–Mariano statistics between 7.6 and 9.9. Consistent with Souto (Souto, 2023), our gain occurred mainly in the asymmetric QLIKE rather than the symmetric MSE-log. Ablations show the result rests neither on a liquidity indicator among the features nor on the nonlinear model's wider feature set, and it is carried in part by persistence entropy, a feature that is invariant to the scale of returns and therefore cannot act as a volatility proxy. When we translate our results into an economic setting, the augmented forecasts deliver VaR coverage closest to nominal and a certainty-equivalent gain of roughly 6 to 9 percent per year for BTC and 17 to 32 percent per year for ETH.

Thus, our experiment supports the notion that the primary benefit of persistent homology comes in turbulent periods and mainly helps nonlinear models. We urge future studies of TDA in financial forecasting to implement controls similar to ours, so as not to overstate the benefit.

Future research building on these findings includes incorporating the persistence image of Definition 7.3 as a high-dimensional input and testing the topological features against an implied-volatility control; extending the construction to joint point clouds over several assets and to degree-two homology; and replicating the experiment on deeper venues and other asset classes with embedding parameters selected by mutual information and false-nearest-neighbour criteria rather than fixed a priori (Section 10.3).

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Statements and Declarations

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Competing Interests. The author has no relevant financial or non-financial interests to disclose.

Data Availability. The one-minute BTC/USDT and ETH/USDT price data underlying this study were obtained from the public Binance US market-data API and are, in principle, re-downloadable by any researcher from that source. The cleaned dataset, derived topological features, and analysis code used to produce the results in this paper are available from the author upon reasonable request; the author intends to archive them in a public repository (e.g., Zenodo) prior to publication.

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